

# test-checklist

## SCORM Package Testing Checklist

### Purpose

This checklist ensures SCORM packages function correctly before deployment to production. Use this for every new package or update.

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### Pre-Upload Testing

#### 1. Package Structure Validation

- ☐ **ZIP file created properly**
  - File extension is .zip
  - No nested ZIP files
  - No hidden files (like .DS\_Store on Mac)
- ☐ **imsmanifest.xml present**
  - Located at root of ZIP (not in subfolder)
  - Valid XML (no syntax errors)
  - Correct SCORM 1.2 schema references
- ☐ **Launch file path correct**
  - File referenced in manifest actually exists
  - Path matches exactly (case-sensitive!)
  - No spaces in filename (use hyphens or underscores)
- ☐ **All resource files included**
  - HTML files
  - JavaScript files (including scorm-wrapper.js)
  - CSS files
  - Images
  - Fonts (if not using CDN)

#### 2. File Size Check

- ☐ **Package size within limits**
  - Total ZIP size: \_\_\_\_\_ MB (check against LMS limit)
  - Largest individual file: \_\_\_\_\_ MB
  - No unnecessarily large files
- ☐ **Image optimization**

- Images compressed appropriately
- No RAW or unoptimized formats
- Reasonable dimensions (no 5000px images)

### 3. Local Testing (Before Upload)

- ☐ **Open launch file in browser**
    - File loads without errors
    - No missing images or broken links
    - Console shows no critical errors
    - SCORM wrapper loads (check console for “[SCORM]” messages)
  - ☐ **Test all interactions**
    - Buttons clickable
    - Drag-and-drop works (if applicable)
    - Forms submit properly
    - Feedback displays correctly
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## Upload and Import Testing

### 4. Package Import

- ☐ **Upload to LMS staging environment**
  - Package imports without errors
  - No timeout during upload
  - No error messages
- ☐ **Package appears in course list**
  - Title displays correctly
  - Description shows (if provided)
  - Thumbnail/icon displays (if applicable)

### 5. Launch Testing

- ☐ **Content launches**
  - Click launch button/link
  - Content loads (not blank screen)
  - Loads within acceptable time (< 10 seconds)
  - No browser security warnings
- ☐ **SCORM API connection**
  - Open browser console (F12)
  - Look for “[SCORM] SCORM initialized successfully” message
  - If API not found, content should still display (preview mode)

### 6. SCORM Tracking Verification

- ☐ **Initialization**

Check console for:  
[SCORM] Initializing SCORM connection...

[SCORM] SCORM API found in window  
[SCORM] SCORM initialized successfully

☐ **Lesson status updates**

- Complete an interaction
- Check LMS gradebook/progress
- Status should change from “Not Started” to “Incomplete”

☐ **Score tracking**

- Complete activity with a score
- Check LMS gradebook
- Score appears correctly (0-100 scale)
- Score: Expected \_\_\_\_\_ / Actual \_\_\_\_\_

☐ **Session time tracked**

- Stay in content for known duration (e.g., 5 minutes)
- Close/exit content
- Check LMS for time spent
- Time: Expected ~5 min / Actual \_\_\_\_\_

☐ **Data persistence (suspend\_data)**

- Make progress in content
  - Close without completing
  - Reopen content
  - Progress should be restored
  - Test: \_\_\_\_\_
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## Cross-Browser Testing

### 7. Desktop Browsers

Test in ALL browsers used by your learners:

**Chrome:** - [ ] Version: \_\_\_\_\_ - [ ] Content loads - [ ] Interactions work - [ ] SCORM tracking works - [ ] No console errors - Issues: \_\_\_\_\_

**Firefox:** - [ ] Version: \_\_\_\_\_ - [ ] Content loads - [ ] Interactions work - [ ] SCORM tracking works - [ ] No console errors - Issues: \_\_\_\_\_

**Safari (Mac):** - [ ] Version: \_\_\_\_\_ - [ ] Content loads - [ ] Interactions work - [ ] SCORM tracking works - [ ] No console errors - Issues: \_\_\_\_\_

**Microsoft Edge:** - [ ] Version: \_\_\_\_\_ - [ ] Content loads - [ ] Interactions work - [ ] SCORM tracking works - [ ] No console errors - Issues: \_\_\_\_\_

**Internet Explorer 11** (if required - note: deprecated): - [ ] Version: 11 - [ ] Content loads (may need polyfills) - [ ] Interactions work - [ ] SCORM tracking works - Issues:

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## 8. Mobile Testing

If mobile support required:

**iOS Safari (iPhone):** - [ ] iOS version: \_\_\_\_\_ - [ ] Content loads - [ ] Responsive design works - [ ] Touch interactions work - [ ] SCORM tracking works - [ ] Orientation works (portrait & landscape) - Issues: \_\_\_\_\_

**iOS Safari (iPad):** - [ ] iOS version: \_\_\_\_\_ - [ ] Content loads - [ ] Responsive design works - [ ] Touch interactions work - [ ] SCORM tracking works - Issues: \_\_\_\_\_

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**Android Chrome:** - [ ] Android version: \_\_\_\_\_ - [ ] Content loads - [ ] Responsive design works - [ ] Touch interactions work - [ ] SCORM tracking works - Issues: \_\_\_\_\_

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## 9. Screen Size Testing

- ☐ **Desktop (1920×1080)**
    - Layout looks good
    - No horizontal scroll
    - Text readable
  - ☐ **Laptop (1366×768)**
    - Layout looks good
    - No horizontal scroll
    - Text readable
  - ☐ **Tablet (768×1024)**
    - Responsive design activates
    - Touch-friendly buttons
    - Readable text size
  - ☐ **Mobile (375×667)**
    - Single column layout
    - Large touch targets
    - No pinch-zoom required for reading
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## Accessibility Testing

### 10. Keyboard Navigation

- ☐ **Tab key navigates through elements**
  - All interactive elements reachable
  - Tab order is logical
  - Focus indicators visible
- ☐ **Keyboard shortcuts work**
  - Enter/Space activates buttons
  - Arrow keys work (if applicable)

- Escape closes modals (if applicable)
- ☐ **No keyboard traps**
  - Can tab out of all components
  - Can exit modals with keyboard

## 11. Screen Reader Testing

Test with at least one screen reader:

**NVDA (Windows - Free):** - [ ] Screen reader detects content - [ ] Headings announced correctly - [ ] Buttons/links labeled properly - [ ] Form fields have labels - [ ] ARIA labels working - Issues: \_\_\_\_\_

**VoiceOver (Mac/iOS - Built-in):** - [ ] Screen reader detects content - [ ] Headings announced correctly - [ ] Buttons/links labeled properly - [ ] Form fields have labels - [ ] ARIA labels working - Issues: \_\_\_\_\_

**JAWS (Windows - Paid):** - [ ] Screen reader detects content - [ ] Headings announced correctly - [ ] Buttons/links labeled properly - [ ] Form fields have labels - [ ] ARIA labels working - Issues: \_\_\_\_\_

## 12. Visual Accessibility

- ☐ **Color contrast sufficient**
    - Text on background meets WCAG AA (4.5:1)
    - Interactive elements meet WCAG AA (3:1)
    - Use WebAIM contrast checker: <https://webaim.org/resources/contrastchecker/>
  - ☐ **Text resizable**
    - Zoom browser to 200%
    - Text remains readable
    - Layout doesn't break
    - No text truncation
  - ☐ **Does not rely on color alone**
    - Correct/incorrect indicated by icons + color
    - Links distinguishable without color
  - ☐ **No flashing content**
    - Nothing flashes more than 3 times per second
    - Animations can be paused
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## Functional Testing

### 13. Learning Interactions

Test every interaction type in the module:

**Multiple Choice:** - [ ] Questions display - [ ] Radio buttons/checkboxes work - [ ] Submit button works - [ ] Feedback displays - [ ] Score calculated correctly

**Drag and Drop:** - [ ] Items draggable (mouse) - [ ] Items draggable (touch on mobile) - [ ] Drop zones accept items - [ ] Snap-to-grid works (if applicable) - [ ] Reset button works - [ ] Check answer works - [ ] Feedback displays

**Matching:** - [ ] Items connect correctly - [ ] Visual feedback works - [ ] Check answer works - [ ] Scoring correct

**Text Input:** - [ ] Input fields accept text - [ ] Validation works - [ ] Submit button works - [ ] Feedback displays

**Hotspot:** - [ ] Image loads - [ ] Clickable areas work - [ ] Hover states work (desktop) - [ ] Feedback displays - [ ] All hotspots accessible

## 14. Navigation

- ☐ **Internal navigation**
  - Next/previous buttons work
  - Menu navigation works (if applicable)
  - Breadcrumbs work (if applicable)
- ☐ **External links**
  - Links open in new tab (best practice)
  - Links work correctly
  - No broken links
- ☐ **Exit behavior**
  - Close button works (if applicable)
  - Browser back button doesn't break content
  - Exit saves progress

## 15. Multimedia

**Images:** - [ ] All images load - [ ] No broken image icons - [ ] Alt text present (check HTML) - [ ] Appropriate file sizes

**Video (if applicable):** - [ ] Video player loads - [ ] Video plays - [ ] Controls work (play, pause, volume, fullscreen) - [ ] Captions available (if required) - [ ] Mobile playback works

**Audio (if applicable):** - [ ] Audio player loads - [ ] Audio plays - [ ] Controls work - [ ] Transcript available (if required) - [ ] Mobile playback works

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## Edge Cases and Error Handling

### 16. Stress Testing

- ☐ **Rapid clicking**
  - Click buttons quickly/repeatedly
  - No JavaScript errors
  - No duplicate submissions
- ☐ **Invalid inputs**
  - Enter unexpected data in forms

- Leave required fields empty
  - Enter very long text
  - Validation handles gracefully
- ☐ **Multiple attempts**
  - Complete activity multiple times
  - Scores update correctly
  - No data corruption

## 17. Connection Issues

- ☐ **Slow connection simulation**
  - Throttle network in browser DevTools
  - Content still loads (may be slow)
  - No broken functionality
- ☐ **Connection interruption**
  - Disconnect network mid-session
  - Reconnect
  - Data saved correctly (check suspend\_data)

## 18. Browser Refresh

- ☐ **F5 refresh during activity**
    - Data preserved (if suspend\_data implemented)
    - OR activity restarts gracefully
    - No errors
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# Performance Testing

## 19. Load Time

- ☐ **Initial load time**
  - Clear cache
  - Load content
  - Time to interactive: \_\_\_\_\_ seconds
  - Target: < 5 seconds on standard connection
- ☐ **Resource loading**
  - Check Network tab in DevTools
  - No 404 errors (missing files)
  - No excessively large files
  - Images lazy-load (if applicable)

## 20. Memory Usage

- ☐ **Memory leaks**
    - Open content
    - Interact extensively
    - Check Task Manager / Activity Monitor
    - Memory usage stable (not constantly increasing)
-

# Security Testing

## 21. Security Checks

- ☐ **No console warnings**
    - No mixed content warnings (HTTP in HTTPS)
    - No security warnings
  - ☐ **Input sanitization**
    - Try XSS attack in text inputs: `<script>alert('test')</script>`
    - Input should be sanitized (not executed)
  - ☐ **No sensitive data exposed**
    - Check browser DevTools > Sources
    - No API keys visible
    - No credentials in code
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# Completion Testing

## 22. Completion Scenarios

**Scenario 1: Full completion** - [ ] Complete all activities - [ ] Check answer and get 100% - [ ] Status in LMS: “Completed” - [ ] Score in LMS: 100

**Scenario 2: Partial completion** - [ ] Complete some activities - [ ] Exit without finishing - [ ] Status in LMS: “Incomplete” - [ ] Progress saved (if applicable)

**Scenario 3: Failed attempt** - [ ] Intentionally get wrong answers - [ ] Status in LMS: Updates appropriately - [ ] Low score recorded

**Scenario 4: Multiple attempts** - [ ] Complete activity (attempt 1) - [ ] Relaunch and complete again (attempt 2) - [ ] Verify which attempt LMS records - [ ] LMS records: [ ] First [ ] Last [ ] Highest [ ] Other

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# Documentation Check

## 23. User-Facing Documentation

- ☐ **Instructions clear**
  - Learner knows what to do
  - No jargon or unclear terms
- ☐ **Feedback messages helpful**
  - Correct/incorrect feedback clear
  - Hints useful (if applicable)
- ☐ **Error messages user-friendly**
  - No technical jargon
  - Tells user what to do next



## 24. Technical Documentation

- ☐ **Module metadata complete**
    - Title correct in manifest
    - Description present
    - Duration estimate accurate
  - ☐ **Known issues documented**
    - Any workarounds noted
    - Limitations explained
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## Sign-Off

### Testing Complete

**Package Name:** \_\_\_\_\_

**Version:** \_\_\_\_\_

**Tested By:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Environment:** - LMS: \_\_\_\_\_ - LMS Version:

\_\_\_\_\_ - Browsers Tested:

\_\_\_\_\_

### Results Summary

**Total Tests:** \_\_\_\_\_ **Passed:** \_\_\_\_\_ **Failed:** \_\_\_\_\_ **Needs Fix:** \_\_\_\_\_

### Critical Issues Found

- 
1. ◦ Severity: [ ] Critical [ ] High [ ] Medium [ ] Low  
◦ Action: \_\_\_\_\_

- 
2. ◦ Severity: [ ] Critical [ ] High [ ] Medium [ ] Low  
◦ Action: \_\_\_\_\_

- 
3. ◦ Severity: [ ] Critical [ ] High [ ] Medium [ ] Low  
◦ Action: \_\_\_\_\_

### Recommendation

- ☐ **Approve for production** - No critical issues
- ☐ **Approve with minor fixes** - Non-blocking issues to address
- ☐ **Reject - needs fixes** - Critical issues must be resolved

**Approver Signature:** \_\_\_\_\_

**Date:** \_\_\_\_\_

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## Quick Reference: Common Issues

| Symptom                     | Likely Cause                 | Solution                         |
|-----------------------------|------------------------------|----------------------------------|
| Blank screen after launch   | Wrong launch file path       | Check manifest, verify file path |
| SCORM API not found         | LMS iframe issue             | Verify SCORM wrapper code        |
| Score not saving            | LMSCcommit not called        | Check SCORM wrapper calls        |
| Content doesn't fit screen  | Missing viewport meta tag    | Add <meta name="viewport" ...>   |
| Images not loading          | Wrong file path              | Use relative paths, check case   |
| Works locally, fails in LMS | Absolute paths used          | Change to relative paths         |
| Mobile touch not working    | Missing touch event handlers | Add touch events to code         |
| Console errors              | JavaScript bugs              | Debug in browser DevTools        |

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